

UWB-IMU-IE: A Full-Trajectory UWB-Inertial Post-Processing System with Recoverability-Guided NLOS Handling

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T01 working-draft status. The equations, proposition, proof, four-stage method, and planned evaluation are carried over from the frozen structure. T00 has verified only the legacy backend build and one repeatable legacy run. The NLOS module, paper runner, locked calibration, and RQ1-RQ4 experiments remain unimplemented or unrun.

Abstract

Offline UWB-inertial post-processing can exploit future measurements to revise historical estimates, but it also needs explicit control of calibration, robustness, uncertainty diagnostics, and evaluation provenance. This T01 working draft defines the planned UWB-IMU-IE system and its evidence contract. The frozen NLOS method uses nonnegative temporally structured support discovery, an unregularized constrained segment refit, candidate-excluded marginal information, and a frozen group-level use-or-suppress rule. Accepted bias states remain in the intended final joint graph rather than becoming exact corrections. At this revision, only the legacy backend build and one repeatable legacy run have been verified; the NLOS implementation and RQ1-RQ4 results are explicitly pending. **[TO COMPLETE: Replace this status text only with locked implementation and experiment evidence; do not invent results.]**

1 Introduction

Start from the task, not a feature inventory. The target is accurate, inspectable *offline* reconstruc-

tion from raw UWB and IMU recordings. The defining system capability is full-trajectory relinearization: a later clean observation can influence an earlier obstructed interval. Dataset adapters, calibration records, diagnostics, and run manifests make this capability testable rather than merely adding software features.

One technical obstacle. Persistent NLOS can corrupt several links simultaneously. Suppression avoids explicitly trusting a biased range but may weaken useful geometric constraints. Conversely, fitting an excess-range variable is not sufficient evidence that its value is reliable: navigation error and bias may be locally interchangeable, and a selected temporal model may be wrong. Existing bias estimators and offline backends must be acknowledged [3–5].

Focused question. *How can a full-trajectory UWB-inertial post-processing system decide when to retain a structured NLOS-bias estimate, using local separability and model-fit evidence, and what changes when future clean context becomes available?* This is not a claim of guaranteed safe recovery or of a universal obstruction-duration threshold.

Intended contributions, pending evidence.
C1—System capability. The intended claim is a reproducible full-trajectory raw-range UWB-inertial pipeline with interchangeable processing policies and graph-consistent diagnostic outputs. T00 currently supports only the narrower legacy build/run fact.
C2—Focused processing method. The frozen

design is a segment-level NLOS module combining adopted support discovery, debiased refitting, candidate-excluded marginal-information assessment, and frozen use/suppress decisions with uncertainty-preserving final inference. It is not implemented at T01. **C3—Evidence and artifact.** The planned evaluation separates system portability, empirical decision quality, multi-link NLOS performance, and future-context effects, with traceable commands and configurations. Those experiments are not run at T01.

Claim boundary. No novelty is claimed for IMU preintegration, tight coupling, robust kernels, nonnegative bias variables, temporal regularization, Schur complementation, or batch optimization individually. The contribution must be demonstrated at the system-capability and validated processing-policy levels.

2 Related Work and Positioning

2.1 Robust estimation and explicit bias handling

Writing target: One paragraph on robust losses/rejection, one on explicit bias estimation. IMU preintegration is an adopted component [1]. The track-constrained UWB/IMU method of Tan et al. and the UWB-only HBR-PGD framework of Yu et al. are close bias-handling references [3, 4]. The decoupled UWB backend is relevant to offline ranging-error refinement [5]. Do not equate “offline experiments” with full-trajectory temporal smoothing or with released code.

2.2 What this paper must distinguish

The intended distinction is a *specific latent bias block* evaluated after accounting for navigation uncertainty, rather than a generic residual score or undifferentiated local curvature. Candidate measurements are absent from the reference graph and then enter the joint block exactly once. This is *not* a held-out-data estimator or an independence guarantee. HBR-PGD-style information weighting must be discussed accurately; a

trace-of-curvature comparator is a diagnostic baseline, not a reimplement of that complete method.

2.3 System positioning and source audit

Writing target: One compact paragraph on verified open estimators and dataset pipelines. Move the full capability matrix to the repository/supplement. Retain evidence columns for raw UWB/IMU, time horizon, explicit bias, diagnostic definition, and artifact availability. Mark unverified cells “not established from reviewed material”, not “No”. **[TO COMPLETE: Read complete closest methods and identify one verified open-system comparator; the supplied discussion is not a substitute for primary-source verification.]**

3 UWB-IMU-IE System

3.1 Scope and interfaces

The main experiments require surveyed anchors, a calibrated lever arm, synchronized clocks, and an independent LOS calibration of static link bias. These prerequisites are not yet established for the current local data. Optional estimated calibration states remain system extensions, not headline contributions. No real-time guarantee, fixed-lag implementation, new sensor integration, or full UAV validation is required by this paper.

The contracted paper-path interface, to be implemented after T01, is *recording + calibration + configuration* $\rightarrow$ *final graph/state + diagnostics + manifest*. It will record input and configuration hashes, software revision, seeds, units/frames, keyframe policy, solver settings, candidate/support masks, accepted groups, calibration provenance, and output status. Calibration and uncertainty assumptions are part of the experiment, not hidden preprocessing. This interface is a contract, not a claim that the roadmap’s suggested runner or metadata types already exist.

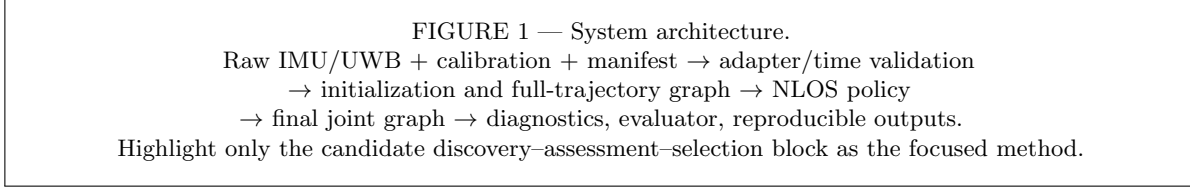


Figure 1: Planned architecture figure. Existing estimator components, the focused NLOS module, and the reproducibility interface should be visually distinct.

3.2 State, measurements, and base graph

At keyframe k , let

$$\mathbf{x}_k = (\mathbf{R}_k, \mathbf{p}_k, \mathbf{v}_k, \mathbf{b}_k^a, \mathbf{b}_k^g), \quad \mathcal{X} = \{\mathbf{x}_k\}_{k=0}^N. \quad (1)$$

The calibrated antenna lever arm is ℓ ; anchor m is at $\mathbf{a}_m$. The range model is

$$h_{km}(\mathcal{X}) = \|\mathbf{p}_k + \mathbf{R}_k \ell - \mathbf{a}_m\|_2, \quad (2)$$

$$z_{km} = h_{km}(\mathcal{X}) + \beta_m + b_{km} + \epsilon_{km}, \quad b_{km} \geq 0. \quad (3)$$

The signed static offset β_m is distinct from dynamic nonnegative excess path b_{km} . The complete range residual can be negative. IMU biases use $\mathbf{b}^a, \mathbf{b}^g$; segment amplitudes will use c_s , avoiding ambiguous notation.

The base objective combines gauge/initial-state priors, IMU preintegration factors, and raw-range factors. Measurements are associated with their actual timestamps through the declared keyframe/interpolation rule; nearest-keyframe approximation, if used, must be documented and bounded. Ground truth is used only for independent calibration/evaluation, never as a hidden navigation factor.

3.3 Validated capabilities, not configuration flags

At T00, the current C++/GTSAM package built successfully in the current workspace. The existing suite executed 38 tests with two failures in the IMU preintegration gravity-parameter semantics. The legacy SFUISE Walk1 path ran twice in isolated directories;

both runs produced byte-identical trajectory, ground-truth, and calibration outputs and an aligned ATE RMSE of 0.169642 m. This is evidence for that legacy path only. It does not validate a paper runner, the NLOS stages, graph-consistent residual/covariance output, raw-frame ATE, simulation, controlled UGV NLOS, or another public-data adapter.

3.4 Artifact contract

The intended release contains source, build environment, tests, one example dataset/configuration, base-line commands, masks, manifests, and scripts regenerating the reported results. At T01, the two contracts, this working draft, the claim–evidence ledger, and T00 logs exist locally; no v2 release artifact exists. Distinguish *implemented*, *tested*, and *released*. **[TO COMPLETE: Insert the actual anonymous repository/archive only after it exists; restore “open-source” to the title only when supported.]**

4 Recoverability-Guided NLOS Processing

This section preserves the frozen method definition. Its equations and proposition specify the implementation target; they are not evidence that Stages 1–4 are present in the repository. The implementation status and unresolved parameters are tracked in the T01 method contract.

4.1 Stage 1: adopted support discovery

Write $J_0 = J_{\text{prior}} + J_{\text{IMU}} + J_{\beta}$ and $w_i = \sigma_i^{-2}$, with $i = (k, m)$. Discover candidate support using

$$\min_{\mathcal{X}, \beta, \mathbf{b} \geq 0} J_0 + \frac{1}{2} \sum_{i \in \mathcal{E}} w_i (h_i + \beta_{m(i)} + b_i - z_i)^2 + \lambda_1 \|\mathbf{b}\|_1 + \lambda_{\text{TV}} \|D\mathbf{b}\|_1. \quad (4)$$

Here D differences consecutive valid samples *within each link*. Split chains at packet gaps exceeding a fixed threshold; specify time/gap conventions. Freeze activity thresholds, minimum segment sample count/duration, and change-point rules using validation data. These choices determine the model and must not be tuned on test labels.

Static-bias reference. Without a reference, $\beta'_m = \beta_m - d$ and $b'_{km} = b_{km} + d$ preserve predictions when feasible. In the noiseless known-trajectory limit, imposing $\min_k b_{km} = 0$ selects $\beta_m = \min_k (z_{km} - h_{km})$; merely assuming that an unknown true LOS epoch exists does not make an unconstrained optimization unique. The main configuration instead fixes β_m from independent LOS calibration. This ambiguity is a short motivation, not a headline theorem.

4.2 Stage 2: debiased segment refit

Let $\hat{\mathcal{S}} = \{\mathcal{S}_s\}$ be disjoint per-link constant-amplitude segments, and let $\mathcal{C}$ be their union of measurements. Use $b_i = c_s \geq 0$ for $i \in \mathcal{S}_s$ and $b_i = 0$ otherwise. Refit $\mathcal{X}, \beta, \mathbf{c}$ without either the ℓ_1 amplitude penalty or TV jump penalty, keeping the segment partition fixed. This removes penalty-induced shrinkage *conditional on the selected partition*; it does not remove support-selection error or segment-model mismatch.

Positive fitted amplitudes are analyzed as interior variables. Boundary or too-short segments are flagged as insufficient evidence, rather than assigned spuriously exact Gaussian uncertainty. The minimum-count rule also prevents one-sample segments from always passing a zero post-fit residual check.

4.3 Stage 3a: a common candidate-excluded reference

Form $\mathcal{G}_0$ from priors, IMU factors, and noncandidate range factors, excluding *all* ranges in $\mathcal{C}$. The term “noncandidate” does not assert true LOS; missed NLOS remains a failure mode. Reference masks and any nonnegative robust weights are fixed and logged at the scoring linearization; the information proxy is the positive-semidefinite Gauss–Newton quadratic, not the exact nonconvex robust Hessian.

Group temporally overlapping segments by connected components of their interval-overlap graph. For a group $\mathcal{A}$, add only its own candidate ranges to $\mathcal{G}_0$ and retain its amplitudes $\mathbf{c}_{\mathcal{A}}$. This conservative reference prevents one group from borrowing confidence from another group that the policy later suppresses. Its score is not an if-and-only-if test of recoverability in the full graph with every candidate retained.

Let $\delta \mathbf{y}$ include *all uncertain nuisance states*: pose, velocity, IMU biases, and static β if estimated. Do not hold these states fixed when claiming to account for their uncertainty. At the debiased linearization, write the whitened perturbation model

$$\begin{bmatrix} \mathbf{J}_{\text{ref}} \\ W_{\mathcal{A}}^{1/2} H_y \end{bmatrix} \delta \mathbf{y} + \begin{bmatrix} 0 \\ W_{\mathcal{A}}^{1/2} H_c \end{bmatrix} \delta \mathbf{c}_{\mathcal{A}} = F \delta \mathbf{y} + G \delta \mathbf{c}_{\mathcal{A}}. \quad (5)$$

$\mathbf{J}_{\text{ref}}$ is the reference Jacobian and H_c is the segment-incidence matrix. No independent amplitude prior is used in the main refit. In particular, discarded ℓ_1/TV penalties must not reappear as fictitious Gaussian precision. Levenberg–Marquardt damping is also not measurement information.

4.4 Stage 3b: marginal information and its limits

Define

$$A = F^{\top} F, \quad B = F^{\top} G, \quad \mathbf{N}_{\mathcal{A}} = G^{\top} G, \quad P_F = F F^{\dagger}. \quad (6)$$

With nonempty disjoint segments and positive retained weights, $\mathbf{N}_{\mathcal{A}} \succ 0$. The rank-aware reduced information is

$$\boxed{\mathbf{R}_{c, \mathcal{A}} = G^{\top} (I - P_F) G.} \quad (7)$$

When $A \succ 0$, this is the usual Schur complement

$$\mathbf{R}_{c,\mathcal{A}} = \mathbf{N}_{\mathcal{A}} - B^\top A^{-1} B. \quad (8)$$

In a proper local Gaussian model this is *marginal precision* of the amplitudes after eliminating $\mathbf{y}$. In contrast, $\mathbf{N}_{\mathcal{A}}$ is their precision *conditional on known nuisance states* [2]. If F is rank deficient, Eq. (7) remains a profile-information quadratic on identifiable directions; it does not by itself define a normalized posterior over unconstrained nuisance states. Use rank-revealing QR/SVD or equivalent sparse elimination, not dense projector formation.

Proposition 1 (Local segment separability in the reference-plus-group model). *For the fixed-support linearized model,*

$$\delta \mathbf{c}^\top \mathbf{R}_{c,\mathcal{A}} \delta \mathbf{c} = \min_{\delta \mathbf{y}} \|F \delta \mathbf{y} + G \delta \mathbf{c}\|_2^2. \quad (9)$$

Thus $\mathbf{R}_{c,\mathcal{A}} \succ 0$ exactly when no nonzero amplitude perturbation can be absorbed by nuisance perturbations with zero local cost. Also $0 \preceq \mathbf{R}_{c,\mathcal{A}} \preceq \mathbf{N}_{\mathcal{A}}$.

Proof. Least-squares elimination projects $G \delta \mathbf{c}$ onto the orthogonal complement of $\text{col}(F)$. Since $I - P_F$ is an orthogonal projector, the identity and both inequalities follow. $\square$

The normalized separability score is

$$\eta_{\mathcal{A}} = \lambda_{\min}(\mathbf{N}_{\mathcal{A}}^{-1/2} \mathbf{R}_{c,\mathcal{A}} \mathbf{N}_{\mathcal{A}}^{-1/2}) \in [0, 1]. \quad (10)$$

It measures the *fraction of nominal bias information retained*, not the absolute accuracy of a correction. Define a companion absolute uncertainty scale from the same matrix:

$$s_{\mathcal{A}} = \begin{cases} [\lambda_{\min}(\mathbf{R}_{c,\mathcal{A}})]^{-1/2}, & \mathbf{R}_{c,\mathcal{A}} \succ 0, \\ +\infty, & \text{otherwise.} \end{cases} \quad (11)$$

Because every c_s is in metres, $s_{\mathcal{A}}$ is in metres; for a proper Gaussian it is the worst-direction standard deviation. No inverse/pseudoinverse trace is used to hide a null direction. For example, $\mathbf{R}_c = \mathbf{N} = 10^{-4} \text{m}^{-2}$ gives $\eta = 1$ but $s = 100 \text{m}$.

Simultaneous links and inertial conditioning. Stacking overlapping amplitudes in Eq. (7) tests their *joint* weakest combination, rather than counting surviving anchors. Geometry enters H_y ; inertial conditioning enters $\mathbf{J}_{\text{ref}}$ together with retained ranges and genuine priors. Component information matrices add before eliminating shared nuisance states, not generally after independent Schur reductions. A full IMU observability theorem is not needed for this claim.

4.5 Stage 3c: empirical model-fit evidence

With nominal, independently calibrated σ_i , use the deliberately simple score

$$\gamma_s = \frac{1}{n_s} \sum_{i \in \mathcal{S}_s} \left(\frac{h_i + \beta_{m(i)} + \hat{c}_s - z_i}{\sigma_i} \right)^2, \quad n_s = |\mathcal{S}_s|. \quad (12)$$

This is average squared normalized *post-fit* residual, not reduced chi-square, NIS, or an independent validation sample. It depends on sample count, leverage, and selected partition. Thresholds are calibrated empirically using separate recordings/seeds and inspected by segment length. Do not whiten by residual-inflated covariance or adaptive robust weights that would automatically conceal poor fit.

Passing γ_s does not establish that a constant-bias model is correct: slowly varying bias can be absorbed by trajectory error or selected into short segments. The main scope therefore retains an empirical check and explicitly measures its failures. A lag-one residual correlation diagnostic is optional supplementary analysis, not an additional default gate.

4.6 Stage 4: frozen group decisions and final joint inference

For an eligible overlap group, the main policy is

$$\text{USE}(\mathcal{A}) \iff \eta_{\mathcal{A}} \geq \tau_\eta \wedge s_{\mathcal{A}} \leq \tau_s \wedge \max_{s \in \mathcal{A}} \gamma_s \leq \tau_\gamma. \quad (13)$$

A rank failure or inadequate segment support fails the rule. Groups are accepted or suppressed as units; partial re-selection within a jointly scored group is not

silently substituted. τ_s expresses the correction-error tolerance in metres; all thresholds are fixed before test evaluation.

Freeze partitions and use/suppress decisions, *not* the accepted amplitudes. With $\mathcal{U}$ the accepted groups, optimize

$$\min_{\mathcal{X}, \beta, c_{\mathcal{U}} \geq 0} J_{\mathcal{G}_0} + \frac{1}{2} \sum_{\mathcal{A} \in \mathcal{U}} \sum_{i \in \mathcal{A}} w_i (h_i + \beta_{m(i)} + c_{s(i)} - z_i)^2. \quad (14)$$

Suppressed candidate ranges are absent; each accepted raw range appears once. Do not also insert its corrected pseudo-range or use a same-data point bias estimate as an independent exact calibration. Keeping c_s preserves bias-trajectory correlations and shared-segment uncertainty.

At the final linearization, audit the same reference-plus-group diagnostics and label them separately from decision-time scores. Export full-final-graph covariances separately when defined. If the final solution is rank-invalid or violates the frozen acceptance checks, perform one conservative fallback with all candidates suppressed and flag the recovery attempt as failed. There is no iterative re-admission loop or general nonlinear convergence guarantee. Report fallback frequency and fallback cost.

4.7 What future context changes

For the *same* historical amplitudes, support, priors, and linearization in a proper Gaussian model, adding a future observation z_f gives the standard identity

$$P_c^+ = P_c - K_{cf} S_f^{-1} K_{cf}^\top \preceq P_c, \quad (15)$$

where $K_{cf} = \text{Cov}(c, z_f \mid \mathcal{D})$ and $S_f = \text{Var}(z_f \mid \mathcal{D}) \succ 0$ [2]. The reduction is nonzero iff $K_{cf} \neq 0$; all amplitude directions improve strictly only if this cross-covariance has full row rank. This is an explanatory identity, not a novel theorem and not a guarantee of lower realized nonlinear estimation error.

Example 1 (Future support under the implemented segment model). *Let a fixed segment mean satisfy $\bar{z} = x + c + e$, with noise precision w , external precision $a > 0$ on the scalar navigation error x , and no prior*

Figure 2: Four-stage NLOS module within UWB-IMU-IE

Input: Validated recording, calibration, frozen configuration

1. Initialize and build the conventional batch graph
2. Discover support using Eq. (4)
3. Freeze segments; refit without ℓ_1 /TV penalties
4. Build common reference $\mathcal{G}_0$ excluding $\mathcal{C}$
5. **for** each eligible overlap group $\mathcal{A}$ **do**
6. Form F, G with full nuisance-state uncertainty
7. Compute $\mathbf{R}_{c, \mathcal{A}}, \eta_{\mathcal{A}}, s_{\mathcal{A}}, \{\gamma_s\}$
8. Freeze group decision using Eq. (13)
- end for**
9. Jointly refit accepted amplitudes and navigation states
10. Audit final graph; use suppression fallback if required
11. Export final states, masks, both score sets, and manifest

on constant amplitude c . Let valid future information increase the external precision by $f(H) \geq 0$. Then

$$\eta(H) = \frac{a + f(H)}{w + a + f(H)}, \quad P_c(H) = \frac{1}{w} + \frac{1}{a + f(H)}. \quad (16)$$

Future context helps through its connection to the historical navigation state, with a candidate-noise floor $1/w$. This illustration uses the same constant-amplitude refit, not a newly introduced random-walk bias process.

Table 1: Evidence allocation. Entries are planned, not measured results.

Question	Main evidence	Primary display
RQ1	Three data types; rerun/time/memory audit	Table 2
RQ2	Repeated LOS and sustained multi-link UGV runs	Table 3, Fig. 3
RQ3	Held-out gates, mismatch and low-information cases	Fig. 3
RQ4	Matched prefixes, fixed historical interval	Fig. 4

5 Experimental Evaluation

All tables and figures in this section are explicit plans with unfilled results. No v2 run has been executed at T01, and the legacy aligned ATE reported in the System section is not a v2 main result or an independent raw-frame benchmark.

5.1 Evidence map and data roles

RQ1: system capability asks whether the same backend, evaluation contract, and provenance work across simulation, controlled data, and one public UWB-IMU dataset. **RQ2: end-to-end value** asks whether the selected configuration improves sustained multi-link NLOS without unacceptable LOS degradation. **RQ3: decision validity** asks whether the diagnostics add predictive and downstream value beyond simpler gates on held-out runs. **RQ4: future context** asks how prefix length changes the same historical interval under matched model/tuning. These are distinct claims; public data without trustworthy bias labels validate portability/trajectory estimates, not bias recoverability.

5.2 Controlled UGV protocol and calibration references

Use the planned 5 m × 5 m motion-capture volume and a surveyed eight-anchor deployment. Add two modest motion profiles, for example a low-turn back-and-forth path and a turning loop/figure-eight. They test different observed conditioning; they do not constitute a complete 3-D inertial-excitation study. Report speed/acceleration summaries and run-to-run path variation.

Prioritize LOS, one-link long obstruction, and sustained two-/three-link obstruction. Short bursts can be supplementary. Use repeated runs and paired evaluation of the same recording for each method. Select equal-cardinality anchor subsets from geometry, without looking at estimator errors, to vary conditioning within one deployment. Keep anchor-count ablations separate. Do not call subset selection a full coplanar-versus-3-D deployment experiment.

Define the measured excess-range *reference*, not noise-free bias truth, as

$$b_{km}^{\text{ref}} = z_{km} - h_{km}(\mathcal{X}^{\text{GT}}) - \hat{\beta}_m^{\text{LOS}}. \quad (17)$$

Its uncertainty includes UWB random range noise, motion capture, anchor survey, lever arm, static calibration, and timing errors, with correlations when relevant. Approximate propagation is

$$\sigma_{b,\text{ref}}^2 \simeq \sigma_z^2 + J_q \Sigma_q J_q^\top, \quad (18)$$

where q stacks calibrated geometric/timing quantities. Use separate LOS data for $\hat{\beta}$. Scripted obstruction times are protocol labels, not perfect packet-wise propagation labels. Report transition buffers and near-noise-floor ambiguity. Synthetic latent b^* is the primary bias-error ground truth; real b^{ref} is a noisy agreement check, especially because z_{km} also enters estimation.

5.3 Baselines and a minimal ablation ladder

Use all-range batch; a preselected Huber/Cauchy batch; a fixed rejection policy; structured bias only;

Table 2: System results to fill only from completed runs. Stage-level runtime includes scoring and any fallback.

Data	Run count	Time	RAM	Rerun Δ
Simulation	TBD	TBD	TBD	TBD
Controlled UGV	TBD	TBD	TBD	TBD
Public UWB-IMU	TBD	TBD	TBD	TBD

structured bias plus debiasing; and the complete module. Add two *gate comparators* to the debiased estimator: fit-only and absolute-uncertainty-plus-fit. The latter is essential: the proposed normalized η must justify its value beyond s_A and generic curvature, not only beyond residual magnitude. Include an η -only gate in simulation to expose its scale and model-mismatch limitations.

All gate comparators share the same candidate supports and debiased candidates for the diagnostic experiment, and use the same final joint refit. State when a comparator is not a full prior-method reproduction. Use oracle bias correction/obstruction-label rejection only as labelled references in simulation, not universal performance bounds. Noise and geometry can make dropping every obstructed range suboptimal.

5.4 RQ1: measured system capability

Report actual adapter/calibration compatibility and unsupported fields. Compare runtime/memory as trajectory length grows on subsampled existing recordings; no new collection campaign is needed. Report sparse-elimination strategy, number/size of candidate groups, and scoring overhead, not just optimizer runtime.

5.5 RQ2: primary trajectory result and LOS control

Use raw-frame ATE as primary after an *independently determined* map-to-ground-truth transform. Aligned ATE/RPE, per-axis errors, and bias-agreement summaries belong in the expanded results. Predeclare

Table 3: Compact primary result table. Use paired run-level summaries and intervals; give complete scenario matrices in the artifact.

Method	LOS ATE	NLOS ATE	P95	Fail.
All ranges	TBD	TBD	TBD	TBD
Robust loss	TBD	TBD	TBD	TBD
Rejection	TBD	TBD	TBD	TBD
Structured bias	TBD	TBD	TBD	TBD
+ debias	TBD	TBD	TBD	TBD
+ fit gate	TBD	TBD	TBD	TBD
+ s /fit gate	TBD	TBD	TBD	TBD
Full module	TBD	TBD	TBD	TBD

trajectory failure, recovery fallback, and acceptable LOS degradation separately. Do not average failed runs away.

5.6 RQ3: decision quality is a head-line experiment

Split by complete recording/trajectory/seed, not by temporally adjacent packets. At matched acceptance coverage, compare accepted-bias error and harmful-acceptance frequency for residual-only, generic-curvature, s -plus-fit, and full gates. Define “harmful correction” before testing, using synthetic true bias and a correction-error tolerance; evaluate downstream trajectory impact separately. Segment decisions are correlated, so confidence intervals resample runs/seeds, not individual packets.

Include matched constant bursts and mismatched within-burst ramps. Sweep a small, predeclared set of ramp slopes including nearly flat and clearly varying cases; add one range-noise/observation-count sweep to expose high- η but low absolute information. Report rejected good corrections as well as accepted bad ones. The objective is calibrated trade-offs, not a promise that γ rejects every mismatch.

Run both a *fixed-partition diagnostic* to isolate model mismatch and the actual end-to-end support-discovery pipeline, which may split a ramp into many short segments. Report segment count/length alongside γ . Thresholds and any choice to add a whiteness

check cannot be selected after inspecting the final test sweep.

5.7 RQ4: matched future-context experiment

For a scripted burst ending at t_b , process prefixes ending at $t_b + H$ for $H \in \{0\text{ s}, 2\text{ s}, 5\text{ s}, \text{full}\}$, interpreting “full” as the complete recording, not a numerical duration. Score the same historical burst interval in every run. $H = 0$ means no post-burst clean context, not a strictly causal estimate at every time inside the burst.

The end-to-end prefix run must redo initialization/support discovery/refit using only that prefix and independent calibration. It must not inherit full-trajectory states, supports, thresholds, or future-dependent noise estimates. A separate common-linearization, fixed-partition diagnostic may test Eq. (15); mark it as analysis, not the online pipeline. Empirical nonlinear $\eta(H)$ need not be monotonic if support, geometry linearization, or nominal information changes.

5.8 Minimum evidence and claim fallback

C1 needs an actual reproducible cross-dataset artifact and measured system behavior, not a feature list. C2 needs an incremental held-out improvement over credible simple gates at matched coverage/cost, together with no unacceptable LOS degradation. It does not require η alone to beat every uncertainty statistic in every regime. If η adds no value beyond s/fit , simplify the gate and narrow the method claim instead of forcing the formula into the paper. If even the full gate is unhelpful, retain the diagnostic as exploratory; system positioning alone does not guarantee publishability. C3 needs reproducible future-context effects in at least one predeclared regime, with failures and limitations reported.

6 Discussion and Conclusion

The intended conclusion is an evidence-backed system result: full-trajectory UWB–inertial post-processing with traceable NLOS decisions and final-graph-consistent uncertainty handling. The local score does not certify model truth, unbiasedness, or global nonlinear observability. Limitations include selected-support uncertainty, nuisance calibration errors, missed NLOS in the reference graph, a conservative all-candidate-excluded reference, limited geometry diversity from one anchor deployment, and offline resource requirements. **[TO COMPLETE: Replace this writing target with quantified findings and verified release status.]**

References

- [1] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza, “On-manifold preintegration for real-time visual-inertial odometry,” arXiv:1512.02363, 2015.
- [2] M. A. Paskin, “Thin junction tree filters for simultaneous localization and mapping,” University of California, Berkeley, Tech. Rep. UCB/CSD-02-1198, 2002, revised May 2003, Appendix A.2.
- [3] R.-Z. Tan, J.-F. Chen, and W.-N. He, “NLoS mitigation with propagation reliability estimation for track-constrained UWB/IMU fusion localization,” *Sensors*, vol. 26, no. 15, Art. 4680, 2026, doi: 10.3390/s26154680.
- [4] Z. Yu, Q. Liu, and Y. Qian, “Heteroscedastic bias-robust projected gradient descent for UWB localization in complex indoor environments,” *Sensors*, vol. 26, no. 14, Art. 4578, 2026, doi: 10.3390/s26144578.
- [5] C. Lu, Z. Guo, X. Qi, L. Mo, and L. Liu, “A decoupled backend optimizer for UWB localization in NLOS environments,” *Measurement*, vol. 276, Art. 121494, 2026, doi: 10.1016/j.measurement.2026.121494.

FIGURE 2 — Headline evidence, not only a successful example.
Left: representative sustained multi-link sequence with obstruction masks,
trajectory errors, estimated amplitudes, and use/suppress decisions.
Right: held-out risk-coverage / harmful-acceptance comparison for simple gates
and the full rule, including matched, ramp-mismatch, and low-information cases.
Show uncertainty across runs; include failures and the operating point.

Figure 3: Planned headline figure coupling the main system behavior with evidence for the decision rule. This is the second and final wide float in the main paper.

FIGURE 3 — Historical bias/trajectory error vs
future horizon.
Use the same evaluation interval and matched
configuration.
Report both support changes and fixed-model
diagnostic trends.

Figure 4: Planned future-context result. The claim concerns information access in offline prefixes, not fixed-lag computational performance.